

Problem 4

Part a)

1. Convergence in Distribution: By definition, $X_k \sim \mathcal{U}(\{0, 1, \dots, 9\})$, so its probability mass function is $\mathbb{P}(X_k = j) = \frac{1}{10}$ for $j \in \{0, \dots, 9\}$. The characteristic function of X_k is:

$$\begin{aligned}
 \varphi_{X_k}(u) &= \mathbb{E}[e^{iuX_k}] \\
 &= \sum_{j=0}^9 \mathbb{P}(X_k = j) e^{iuj} \\
 &= \sum_{j=0}^9 \frac{1}{10} e^{iuj} \\
 &= \frac{1}{10} \sum_{j=0}^9 e^{iuj} \quad (\text{geometric sum formula}) \\
 &= \frac{1}{10} \frac{1 - e^{iu10}}{1 - e^{iu}}, \quad \text{for } u \neq 0
 \end{aligned} \tag{1}$$

Because the random variables $X_1, \dots, X_n$ are independent, we have for $Y_n = \sum_{i=1}^n X_i 10^{-i}$:

$$\begin{aligned}
 \varphi_{Y_n}(u) &= \prod_{i=1}^n \varphi_{10^{-i} \cdot X_i}(u) && (\text{by extending Lemma 3.5.6 via induction}) \\
 &= \prod_{i=1}^n \varphi_{X_i}(u 10^{-i}) && (\text{Theorem 3.5.2 b))}) \\
 &= \prod_{i=1}^n \frac{1}{10} \frac{1 - e^{iu 10^{-i} \cdot 10}}{1 - e^{iu 10^{-i}}} && (\text{by (1)}) \\
 &= \frac{1}{10^n} \prod_{i=1}^n \frac{1 - e^{iu 10^{-i+1}}}{1 - e^{iu 10^{-i}}} \\
 &= \frac{1}{10^n} \frac{1 - e^{iu 10^0}}{1 - e^{iu 10^{-n}}} && (\text{telescoping product}) \\
 &= \frac{1 - e^{iu}}{\frac{1 - e^{iu 10^{-n}}}{10^{-n}}}
 \end{aligned}$$

Let $x = 10^{-n}$. As $n \rightarrow \infty$, we have $x \rightarrow 0$. Then, for the denominator, we have:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} 10^n (1 - e^{iu 10^{-n}}) &= \lim_{x \rightarrow 0} \frac{1 - e^{iux}}{x} \\
 &\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{-iue^{iux}}{1} \\
 &= -iu.
 \end{aligned}$$

Therefore, for any $u \neq 0$, the characteristic function converges to:

$$\lim_{n \rightarrow \infty} \varphi_{Y_n}(u) = \frac{1 - e^{iu}}{-iu} = \frac{e^{iu} - 1}{iu}.$$

For $u = 0$, we evaluate the characteristic function directly: $\varphi_{Y_n}(0) = \mathbb{E}[e^0] = 1$ for all $n \in \mathbb{N}$, so $\lim_{n \rightarrow \infty} \varphi_{Y_n}(0) = 1$. To ensure the limit function is continuous at the origin (a requirement for

Lévy's Continuity Theorem), we must show that the limit of the expression for $u \neq 0$ approaches 1 as $u \rightarrow 0$. Since evaluating at $u = 0$ yields a $\frac{0}{0}$ indeterminate form, we apply L'Hôpital's rule:

$$\lim_{u \rightarrow 0} \frac{e^{iu} - 1}{iu} \stackrel{\text{L'H}}{=} \lim_{u \rightarrow 0} \frac{ie^{iu}}{i} = \lim_{u \rightarrow 0} e^{iu} = 1.$$

For $u = 0$, $\varphi_{Y_n}(0) = 1 \rightarrow 1$, which continuously matches $\lim_{u \rightarrow 0} \frac{e^{iu} - 1}{iu} = 1$. We have shown that for every $u \in \mathbb{R}$, the sequence of characteristic functions converges pointwise to a limit function $f(u)$:

$$\varphi_{Y_n}(u) \xrightarrow{n \rightarrow \infty} f(u) := \begin{cases} \frac{e^{iu} - 1}{iu} & \text{if } u \neq 0, \\ 1 & \text{if } u = 0. \end{cases}$$

Let $Y \sim \mathcal{U}([0, 1])$. Its characteristic function is:

$$\varphi_Y(u) = \int_0^1 e^{iux} \cdot 1 \, dx = \left[\frac{e^{iux}}{iu} \right]_0^1 = \frac{e^{iu} - 1}{iu}.$$

Then $\varphi_{Y_n}(u) \xrightarrow{n \rightarrow \infty} \varphi_Y(u)$ for all $u \in \mathbb{R}$, and $\varphi_Y(u)$ is continuous at $u = 0$. By Lévy's Continuity Theorem (Theorem 4.2.4), $Y_n \xrightarrow{d} Y$, and, by the uniqueness theorem (Theorem 3.5.4), the limit distribution is $\mathcal{U}([0, 1])$. □

2. Almost Sure Convergence:

Yes, the sequence converges $\mathbb{P}$ -almost surely. Since $X_k(\omega) \geq 0$ for all $k \in \mathbb{N}$ and all $\omega \in \Omega$, the sequence of partial sums $Y_n(\omega) = \sum_{k=1}^n X_k(\omega) 10^{-k}$ is monotonically increasing for every $\omega \in \Omega$. Also, the sequence is deterministically bounded from above, because the maximum value any X_k can take is 9:

$$Y_n(\omega) \leq \sum_{k=1}^n 9 \cdot 10^{-k} = 9 \cdot \frac{10^{-1} - 10^{-(n+1)}}{1 - 10^{-1}} = 1 - 10^{-n} < 1, \quad \forall n \in \mathbb{N}.$$

Since $(Y_n(\omega))_{n \in \mathbb{N}}$ is a bounded, monotonically increasing sequence of real numbers, it must converge pointwise for $\omega \in \Omega$ (and, hence, $\mathbb{P}$ -almost sure convergence, Definition 3.3.10). Let $Y^*(\omega) = \lim_{n \rightarrow \infty} Y_n(\omega)$, meaning $Y_n \xrightarrow{a.s.} Y^*$. Then, by Theorem 4.1.2 a), $Y_n \xrightarrow{p} Y^*$, and, by Theorem 4.1.8 a), $Y_n \xrightarrow{d} Y^*$.

From the first part of the problem, we already established that $Y_n \xrightarrow{d} Y$, where $Y \sim \mathcal{U}([0, 1])$. Since both $Y_n \xrightarrow{d} Y^*$ and $Y_n \xrightarrow{d} Y$, then $\varphi_{Y^*}(u) = \varphi_Y(u)$. By the uniqueness theorem (Theorem 3.5.4), Y^* and Y share the exact same distribution, meaning $Y^* \sim \mathcal{U}([0, 1])$.

Part a was submitted and got full score, however I did not submit the solution to part b and c because I didn't have time to properly check and clarify certain parts, so I am not sure if it is correct but it seemed to make sense.

Part b)

Step 1. Distribution Function of $\frac{\pi M_n}{n}$ By definition, $X_i \sim \text{Cauchy}(0, 1)$, so its density is $f_X(t) = \frac{1}{\pi(1+t^2)}$. The corresponding distribution function is:

$$F_X(t) = \int_{-\infty}^t \frac{1}{\pi(1+s^2)} \, ds = \frac{1}{\pi} \arctan(t) + \frac{1}{2}.$$

Let $M_n = \max\{X_1, \dots, X_n\}$ and $Y_n := \frac{\pi M_n}{n}$. For this scaled random variable sequence, we define the distribution function shorthand $F_n(x) := F_{Y_n}(x)$:

$$\begin{aligned}
F_n(x) &:= \mathbb{P}\left(\frac{\pi M_n}{n} \leq x\right) \\
&= \mathbb{P}\left(M_n \leq \frac{nx}{\pi}\right) \\
&= \mathbb{P}\left(X_1 \leq \frac{nx}{\pi}, X_2 \leq \frac{nx}{\pi}, \dots, X_n \leq \frac{nx}{\pi}\right) && \text{(definition of maximum)} \\
&= \prod_{i=1}^n \mathbb{P}\left(X_i \leq \frac{nx}{\pi}\right) && \text{(by independence)} \\
&= \left[F_X\left(\frac{nx}{\pi}\right)\right]^n.
\end{aligned} \tag{2}$$

Step 2. Pointwise Convergence of the CDF:

We must evaluate the limit of $F_n(x)$ as $n \rightarrow \infty$ for all $x \in \mathbb{R}$ to apply Theorem 4.1.6. Because the necessary trigonometric identities require positive arguments, we partition the real number line into two cases:

1. *Case 1: $x \leq 0$.*

If $x \leq 0$, then $\frac{nx}{\pi} \leq 0$, which implies $\arctan\left(\frac{nx}{\pi}\right) \leq 0$. Hence, $F_X\left(\frac{nx}{\pi}\right) \leq \frac{1}{2}$.

$$0 \leq F_n(x) = \left[F_X\left(\frac{nx}{\pi}\right)\right]^n \leq \left(\frac{1}{2}\right)^n \xrightarrow{n \rightarrow \infty} 0.$$

2. *Case 2: $x > 0$.*

For $t > 0$, we utilize the trigonometric identity $\arctan(t) + \arctan(1/t) = \frac{\pi}{2}$. Letting $t = \frac{nx}{\pi} > 0$, we rewrite the CDF of X :

$$F_X\left(\frac{nx}{\pi}\right) = \frac{1}{2} + \frac{1}{\pi} \left(\frac{\pi}{2} - \arctan\left(\frac{\pi}{nx}\right)\right) = 1 - \frac{1}{\pi} \arctan\left(\frac{\pi}{nx}\right).$$

Substituting this into (2) yields:

$$\begin{aligned}
F_n(x) &= \left(1 - \frac{1}{\pi} \arctan\left(\frac{\pi}{nx}\right)\right)^n \\
&= \exp\left[n \ln\left(1 - \frac{1}{\pi} \arctan\left(\frac{\pi}{nx}\right)\right)\right].
\end{aligned}$$

To evaluate the limit of the exponent, let $z = \frac{1}{n}$. As $n \rightarrow \infty$, $z \rightarrow 0^+$:

$$\begin{aligned}
\lim_{n \rightarrow \infty} n \ln\left(1 - \frac{1}{\pi} \arctan\left(\frac{\pi}{nx}\right)\right) &= \lim_{z \rightarrow 0^+} \frac{\ln\left(1 - \frac{1}{\pi} \arctan\left(\frac{\pi z}{x}\right)\right)}{z} \\
&\stackrel{\text{L'H}}{=} \lim_{z \rightarrow 0^+} \frac{\left(1 - \frac{1}{\pi} \arctan\left(\frac{\pi z}{x}\right)\right)^{-1} \cdot \left(-\frac{1}{\pi}\right) \cdot \left(1 + \left(\frac{\pi z}{x}\right)^2\right)^{-1} \cdot \frac{\pi}{x}}{1} \\
&= \frac{1 \cdot \left(-\frac{1}{\pi}\right) \cdot 1 \cdot \frac{\pi}{x}}{1} \\
&= -\frac{1}{x}.
\end{aligned}$$

By the continuity of the exponential function, we conclude $\lim_{n \rightarrow \infty} F_n(x) = \exp\left(-\frac{1}{x}\right)$ for $x > 0$.

Combining both cases, we have established that the sequence of distribution functions converges pointwise for all $x \in \mathbb{R}$ to the limit function $F_M(x)$:

$$\lim_{n \rightarrow \infty} F_n(x) = F_M(x) := \begin{cases} \exp(-1/x) & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

The limit function $F_M(x)$ is a valid distribution function and is continuous everywhere on $\mathbb{R}$ (specifically at $x = 0$, since $\lim_{x \rightarrow 0^+} e^{-1/x} = 0$). By Theorem 4.1.6, pointwise convergence of the CDF at all continuity points of the limit CDF implies convergence in distribution. Therefore, $\frac{\pi M_n}{n} \xrightarrow{d} M$.

Part c)

We will prove this utilizing characteristic functions and Lévy's Continuity Theorem (Theorem 4.2.4). Since $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$, we know from Example 3.5.3 (item 5) that its characteristic function is:

$$\varphi_{X_n}(u) = \exp\left(iu\mu_n - \frac{1}{2}\sigma_n^2 u^2\right), \quad u \in \mathbb{R}.$$

We are given that $X_n \xrightarrow{d} X$. By Lévy's Continuity Theorem (Theorem 4.2.4 a), the sequence of characteristic functions $\varphi_{X_n}(u)$ must converge pointwise to $\varphi_X(u)$, where φ_X is the characteristic function of X . Furthermore, $\varphi_X(u)$ must be continuous at $u = 0$ with $\varphi_X(0) = 1$.

Taking the absolute value of the characteristic functions, we get:

$$|\varphi_{X_n}(u)| = \exp\left(-\frac{1}{2}\sigma_n^2 u^2\right) \xrightarrow{n \rightarrow \infty} |\varphi_X(u)|.$$

Since φ_X is continuous at 0 and $\varphi_X(0) = 1$, there must exist a small neighborhood around 0 where φ_X is non-zero. Let $u_0 \neq 0$ be a point in this neighborhood such that $|\varphi_X(u_0)| > 0$. We then have:

$$\exp\left(-\frac{1}{2}\sigma_n^2 u_0^2\right) \xrightarrow{n \rightarrow \infty} |\varphi_X(u_0)| > 0.$$

Taking the natural logarithm (which is continuous on $(0, \infty)$):

$$-\frac{1}{2}\sigma_n^2 u_0^2 \xrightarrow{n \rightarrow \infty} \ln |\varphi_X(u_0)| \implies \sigma_n^2 \xrightarrow{n \rightarrow \infty} -\frac{2}{u_0^2} \ln |\varphi_X(u_0)|.$$

Let $\sigma^2 := -\frac{2}{u_0^2} \ln |\varphi_X(u_0)|$. Since $|\varphi_X(u_0)| \leq 1$, the logarithm is non-positive, meaning $\sigma^2 \geq 0$. Thus, the variance sequence converges to a finite, non-negative limit: $\sigma_n^2 \rightarrow \sigma^2$.

Knowing that $\sigma_n^2 \rightarrow \sigma^2$, we can analyze the phase of the characteristic function. For any $u \in \mathbb{R}$:

$$\varphi_{X_n}(u) = \exp(iu\mu_n) \exp\left(-\frac{1}{2}\sigma_n^2 u^2\right) \xrightarrow{n \rightarrow \infty} \varphi_X(u).$$

Since $\exp(-\frac{1}{2}\sigma_n^2 u^2) \rightarrow \exp(-\frac{1}{2}\sigma^2 u^2)$ and this exponential is strictly greater than zero, we can isolate the phase component:

$$\exp(iu\mu_n) \xrightarrow{n \rightarrow \infty} \frac{\varphi_X(u)}{\exp(-\frac{1}{2}\sigma^2 u^2)} =: g(u). \quad (3)$$

We now show that the sequence of means $(\mu_n)_{n \in \mathbb{N}}$ must also converge to a finite limit. First, we rule out that $|\mu_n| \rightarrow \infty$. By integrating the left side of (3) over a symmetric interval $[-\delta, \delta]$, we

obtain $\int_{-\delta}^{\delta} \exp(iu\mu_n) du = \frac{2 \sin(\delta\mu_n)}{\mu_n}$. If $|\mu_n| \rightarrow \infty$, this integral would converge to 0. However, the integral of the right side, $\int_{-\delta}^{\delta} g(u) du$, does not equal zero for sufficiently small $\delta > 0$ because $g(0) = 1$ and g is continuous at 0. Thus, $(\mu_n)_{n \in \mathbb{N}}$ must be bounded. Let μ and μ' be two limit points of the bounded sequence $(\mu_n)_{n \in \mathbb{N}}$. Then, taking the limit along the corresponding subsequences in (3) gives $\exp(iu\mu) = g(u)$ and $\exp(iu\mu') = g(u)$ for all $u \in \mathbb{R}$. This implies $\exp(iu(\mu - \mu')) = 1$ for all $u \in \mathbb{R}$, which is only mathematically possible if $\mu = \mu'$. Because (μ_n) is bounded and has exactly one limit point, the sequence itself converges to $\mu \in \mathbb{R}$.

Finally, since $\mu_n \rightarrow \mu$ and $\sigma_n^2 \rightarrow \sigma^2$, the characteristic function of X_n converges as follows:

$$\lim_{n \rightarrow \infty} \varphi_{X_n}(u) = \lim_{n \rightarrow \infty} \exp\left(iu\mu_n - \frac{1}{2}\sigma_n^2 u^2\right) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right).$$

By Lévy's Continuity Theorem (Theorem 4.2.4), $\varphi_X(u) = \exp(iu\mu - \frac{1}{2}\sigma^2 u^2)$. This is exactly the characteristic function of a $\mathcal{N}(\mu, \sigma^2)$ distribution (if $\sigma^2 = 0$, it represents the degenerate Dirac measure δ_μ). By the Uniqueness Theorem (Theorem 3.5.4), $X \sim \mathcal{N}(\mu, \sigma^2)$.